Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 08)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Square-mesh escape panels .....

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Minimum and maximum catchable sizes .....

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No-take zones .....

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Closed seasons .....

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0 1 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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0 2 Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km^2	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km^2	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km^2	under 250	over 50% in 10 years or 3 generations

0 2 . 1 State **three** roles of the IUCN.

[3 marks]

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0 2 . 2 A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km^2 of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 2 . 3 Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 2 . 4 Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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0 3 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

0 3 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer _____ million years

0 3 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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0 3 . 3 Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 3 . 4 Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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0 3 . 5 Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

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0 4 . 1 Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

0 4 . 2 Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

0 4 . 3 Describe **three** threats to UK broadleaf woodland.

[3 marks]

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0 5 . 1 Describe how agriculture alters the hydrological cycle.

[3 marks]

0 5 . 2 Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

Organic food

0 5 . 3 Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

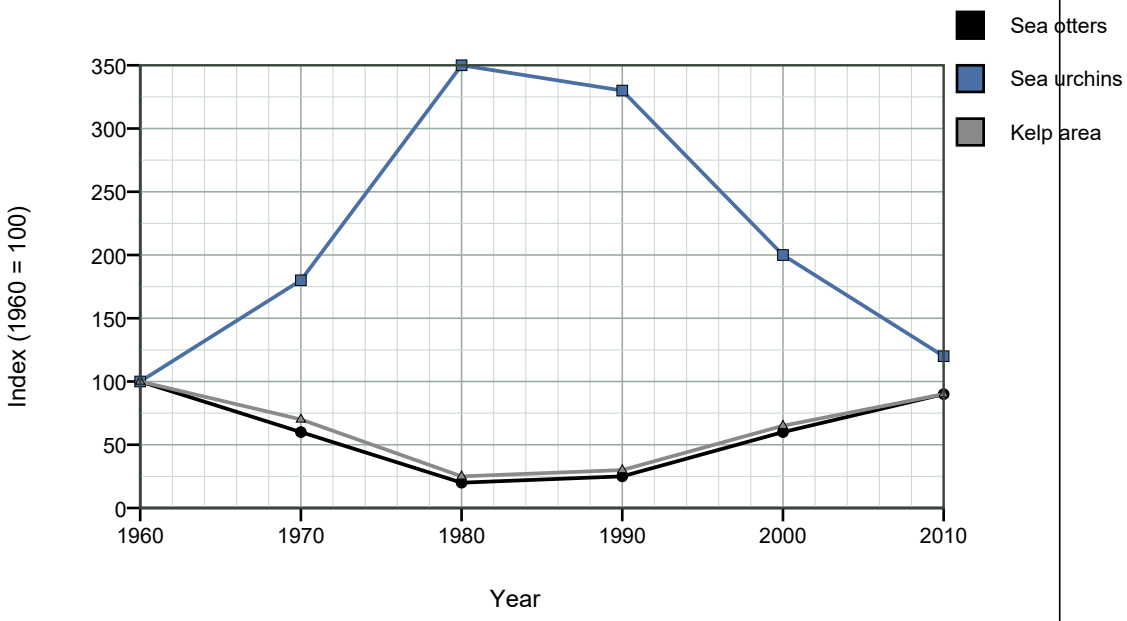
[3 marks]

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0 6 Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 6 . 1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

0 6 . 2 Explain the relationships shown in Figure 1.

[3 marks]

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0 6 . 3 Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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0 6 . 4 Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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0 7 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

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0 7 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

0 7 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

0 7 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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0 8 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 8 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 8 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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5

0 9 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 9 . 1 Explain the main advantage of GM over conventional selective breeding.

[2 marks]

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0 9 . 2 For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops .....

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Roundup Ready soya .....

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Golden rice .....

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0 9 . 3 Explain **four** concerns about GM crops.

[4 marks]

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0 9 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

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10.1 Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **rewilding**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS